



Semester Project Proposal

Semester Project Title: AI Resume Analyzer

Student Details

(Fill according to number of team/group members)

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1. Introduction & Problem Statement

1.1 Problem statement

Traditional resume screening is a time-consuming and subjective process. Recruiters must manually read and interpret each resume to assess its relevance to a job description, leading to:

- Inconsistent evaluations due to human bias.
- Delays in the hiring process.
- Overlooked candidates who may fit the role but lack specific keywords.

On the other hand, applicants lack real-time feedback on whether their resume is well-suited for a particular job posting.

Therefore, there is a need for an automated AI-based system that:

1. Compares resumes with job descriptions using semantic similarity techniques.
2. Provides a quantitative score representing candidate-job match accuracy.
3. Highlights missing or weak skill areas in the resume.
4. Helps both recruiters and candidates make data-driven hiring decisions.

The AI Resume Analyzer addresses these issues by applying NLP and machine learning to automate and enhance the resume screening process, ensuring fairness, speed, and consistency in recruitment.

1.2 Introduction

In today's highly competitive job market, recruiters receive hundreds of resumes for a single position, making it difficult to manually evaluate and shortlist candidates efficiently. Similarly, applicants often struggle to tailor their resumes effectively for specific job descriptions, resulting in missed opportunities.

The **AI Resume Analyzer** aims to bridge this gap by leveraging Artificial Intelligence (AI) and Natural Language Processing (NLP) to automatically analyse resumes and job descriptions. It provides insights on how well a candidate's profile aligns with the job requirements by calculating a *match percentage*, highlighting missing keywords, and suggesting improvements.

This system not only reduces recruiters' screening time but also helps candidates optimize their resumes to increase their chances of selection. By integrating **TensorFlow-based NLP models** with a **Spring Boot** backend and **PostgreSQL** database, the project offers an intelligent, scalable, and user-friendly platform for automated resume evaluation.

2. Objectives of the Project

- Provide a user-friendly web interface for candidates and recruiters.
- Maintain a secure and organized database of user profiles, resumes, and job descriptions.
- Automate the comparison between resumes and job descriptions using AI and NLP techniques.
- Generate a match score and highlight missing or relevant keywords for improvement.
- Enable recruiters to quickly shortlist candidates based on AI-driven insights.
- Enable recruiters to give feedback to applicants
- Ensure data privacy, security, and accuracy throughout the resume analysis process.

3 Requirements Engineering

3.1 Functional Requirements

- **User Registration and Login**
The system shall allow users (candidates and recruiters) to create accounts and log in securely. The system shall verify user credentials before granting access.
- **Resume Upload**
The system shall allow candidates to upload resumes in supported formats (PDF, DOCX, etc.). The system shall extract text content from uploaded resumes for analysis.
- **Job Description Input**
The system shall allow recruiters to enter or upload job descriptions. The system shall store job descriptions for future comparisons.
- **AI-Based Resume Analysis**
The system shall use an AI model (TensorFlow/NLP) to analyse and compare resumes with job descriptions. The system shall calculate a **match percentage** indicating the level of similarity. The system shall identify **missing or weak keywords** in the resume.
- **Result Generation and Feedback**
The system shall display the analysis result in an easy-to-understand format (score, keywords, and feedback). The system shall allow users to download or view detailed reports.
- **Database Management**
The system shall store user profiles, resumes, job descriptions, and analysis results in a PostgreSQL database. The system shall ensure consistency, integrity, and security of stored data.
- **Report Generation**
The system shall generate reports for recruiters showing shortlisted candidates and analysis statistics. The system shall allow candidates to view their resume improvement history.
- **Admin Functions (optional)**
The system shall allow administrators to manage user accounts and system settings. The system shall monitor system performance and data usage.

- **Security and Privacy**
The system shall encrypt sensitive data (e.g., passwords and personal details). The system shall prevent unauthorized access to user data and files.

3.2 Non-Functional Requirements

- **Performance Requirements**
The system shall process and analyse a resume within 5–10 seconds under normal load. The system shall handle multiple simultaneous users without noticeable delays. The database queries shall execute efficiently to minimize response time.
- **Scalability Requirements**
The system shall be designed to accommodate an increasing number of users, resumes, and job descriptions. The architecture shall support easy integration of additional AI models or modules in the future.
- **Usability Requirements**
The system shall provide an intuitive and responsive web interface accessible from both desktop and mobile devices. The system shall use clear visual indicators (progress bars, result summaries, color codes) for better user understanding. The system shall include proper navigation, help sections, and feedback messages.
- **Reliability and Availability**
The system shall ensure that data remains available and consistent even after unexpected failures. The system shall maintain at least 99% uptime during normal operation. Automatic recovery mechanisms shall be implemented for database or server crashes.
- **Security Requirements**
The system shall use secure authentication and authorization protocols. All user data, especially resumes and credentials, shall be encrypted during storage and transmission. The system shall protect against common web vulnerabilities (e.g., SQL injection, XSS).
- **Maintainability Requirements**
The system shall have a modular code structure to allow easy maintenance and updates. Documentation shall be provided for all major components and APIs. The system shall log all major actions and errors for debugging and auditing.
- **Portability Requirements**
The system shall be deployable on various platforms (Windows, Linux, cloud environments). The application shall be accessible through standard web browsers without additional plugins.
- **Data Integrity and Accuracy**
The system shall ensure that uploaded data is not corrupted or lost during processing. AI-generated results shall maintain at least 90% accuracy based on validated test data.